

SAO PAULO CONGONHAS, Brazil

WMO: 837800

Lat: 23.627S

Lon: 46.655W

Elev: 802

StdP: 92.06

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	10.8	3.2	5.2	20.1	5.1	6.0	18.7	8.5	20.8	7.5	20.1	2.9	170	0.331

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	8.2	32.2	20.1	31.2	20.1	30.1	20.1	23.2	27.6	22.5	27.1	22.0	26.8	3.9	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	18.4	25.5	21.2	17.5	24.5	20.8	17.1	24.2	73.2	27.8	70.7	27.1	68.7	26.9	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.1	7.2	6.4	DB	6.9	34.5	2.1	0.9	5.4	35.1	4.1	35.7	2.9	36.2	1.4	36.8	(4)
(5)			WB	5.4	24.5	2.0	1.2	4.0	25.4	2.8	26.1	1.7	26.8	0.3	27.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	21.0	23.9	24.3	23.4	21.8	19.0	18.1	17.8	18.8	19.7	21.0	21.5	23.2	(6)	
	DBStd	3.63	2.41	2.31	2.31	2.54	2.64	2.78	3.13	3.38	3.66	3.31	2.97	2.57	(7)	
	HDD10.0	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)	
	HDD18.3	201	1	0	0	4	24	37	48	37	28	13	8	1	(9)	
	CDD10.0	4022	429	401	414	355	279	244	241	273	292	340	345	408	(10)	
	CDD18.3	1181	172	167	157	109	45	31	31	52	69	94	103	152	(11)	
	CDH23.3	6300	905	966	759	490	167	118	161	354	486	569	519	805	(12)	
(13)	CDH26.7	1796	279	310	212	106	17	5	11	77	175	214	151	241	(13)	
(14)	Wind	WSAvg	3.0	3.0	3.0	3.0	2.9	2.8	2.5	2.6	2.8	3.3	3.4	3.6	3.4	(14)
(15) Precipitation	PrecAvg	1562	282	219	195	83	68	61	57	33	91	119	152	201	(15)	
	PrecMax	2198	604	397	393	163	183	260	160	88	215	202	297	375	(16)	
	PrecMin	1141	156	89	87	13	15	10	8	2	13	23	47	57	(17)	
	PrecStd	244	100	74	65	40	34	54	44	23	55	42	60	67	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	33.2	32.1	30.9	28.2	27.2	28.0	30.7	33.1	33.9	32.1	32.6	(19)	
		MCWB	21.1	21.3	21.1	20.1	18.7	17.7	16.5	16.9	17.8	19.1	19.8	20.6	(20)	
	2%	DB	31.2	31.8	30.7	29.2	26.9	26.0	26.5	28.8	30.9	31.5	30.1	31.0	(21)	
		MCWB	21.1	21.1	21.0	19.8	18.1	17.2	16.0	16.3	17.0	19.1	20.1	20.7	(22)	
	5%	DB	29.8	30.1	29.1	27.9	25.2	24.9	25.1	27.1	28.9	29.1	28.2	29.2	(23)	
		MCWB	21.3	21.3	21.0	19.8	17.7	16.6	15.9	16.1	17.3	19.4	20.2	20.7	(24)	
	10%	DB	28.0	28.8	27.5	26.1	23.8	23.2	23.9	25.2	26.1	26.4	26.2	27.8	(25)	
		MCWB	21.3	21.3	20.9	19.7	17.4	16.2	15.5	15.8	17.4	19.2	19.8	20.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.7	24.4	23.5	22.5	20.4	19.6	18.4	19.7	20.7	22.3	23.2	23.4	(27)	
		MCDB	28.0	29.5	28.1	26.9	24.7	24.6	23.9	25.6	26.4	27.5	26.5	27.4	(28)	
	2%	WB	22.8	23.1	22.5	21.6	19.6	18.4	17.5	18.4	19.6	21.1	21.8	22.3	(29)	
		MCDB	27.3	27.8	27.2	25.8	23.8	23.1	23.0	24.2	25.5	26.5	26.9	27.0	(30)	
	5%	WB	22.2	22.3	21.9	21.0	18.8	17.6	16.9	17.5	18.8	20.3	21.0	21.7	(31)	
		MCDB	26.7	27.1	26.6	24.9	22.8	22.1	22.1	23.2	24.4	25.6	25.6	26.4	(32)	
	10%	WB	21.7	21.8	21.3	20.4	18.2	17.0	16.3	16.7	18.1	19.7	20.3	21.2	(33)	
		MCDB	26.0	26.4	25.5	24.2	22.2	21.4	21.2	22.2	23.1	24.6	24.7	25.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	7.8	8.2	7.8	7.9	7.5	8.2	8.8	9.6	9.1	8.7	8.2	8.2	(35)	
		MCDBR	10.2	10.4	9.8	9.8	9.5	10.2	11.3	12.8	13.4	13.2	11.2	10.4	(36)	
	5% WB	MCWBR	3.8	3.9	3.6	3.6	3.5	3.7	4.0	4.2	4.1	4.4	4.1	3.9	(37)	
		MCWBR	8.7	9.2	8.8	8.4	8.4	9.0	9.7	10.9	10.7	10.8	9.3	9.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.393	0.392	0.383	0.371	0.349	0.353	0.363	0.397	0.488	0.447	0.416	0.400	(40)	
		taud	2.469	2.453	2.483	2.489	2.533	2.502	2.459	2.330	2.053	2.240	2.342	2.420	(41)	
	Ebn at Noon		951	938	919	886	868	840	843	846	807	877	923	946	(42)	
		Edh at Noon	119	119	110	101	89	88	94	117	167	145	135	125	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.26	5.54	4.91	4.40	3.52	3.22	3.38	4.08	4.37	4.91	5.26	5.54	(44)	
	RadStd		0.75	0.48	0.47	0.36	0.27	0.31	0.36	0.36	0.55	0.57	0.32	0.39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-0.38	-0.45	N/A	N/A	N/A	N/A	(46)
(47)	Regional (3 neighbors)	N/A	N/A	N/A	N/A	-0.30	-0.40	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page